

Can Chat GPT be a useful didactic tool in teaching and learning advanced mathematical analysis optimization topics?

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Chat GPT (Chat Generative Pre-trained Transformed) was developed by research organization OpenAI and launched in November 2022. It is available in two online versions: a free version based on GPT-3.5 and a more advanced commercial version based on GPT-4. Based on the LLM (Large Language Model), Chat GPT works by learning statistical language patterns from huge databases of online texts, which also contain various types of false information and outdated knowledge. It allows users to conduct and refine conversations, and subsequent answers, questions and suggestions are treated as context at every stage of the conversation. It can both download and create images, as well as use math and coding software. Chatbots, in general, can generate fabricated information called hallucinations. [5,6,7]

The topic of using artificial intelligence, in particular Chat GPT, in teaching and learning various subjects - including mathematics and computer science - has been undertaken and widely discussed in recent years. [1,2,3,4] Related articles highlight the educational utility of Chat GPT as a comprehensive tool that provides users with access to basic knowledge in mathematics and other fields, including access to basic definitions, methods, instructions and algorithms. But on the other hand, attention was also drawn to Chat GPT's lack of deep understanding of some issues, as well as the lack of effective opportunities to correct its “false beliefs”. Chat GPT has become a popular tool in the last two years, also used by students. At the Faculty of Applied Informatics and Mathematics of Warsaw University of Life Science, a student research club dedicated to Chat GPT has been operating for some time, in which some students participate. This talk discusses the usefulness of Chat GPT as a didactic tool in teaching and learning advanced topics and optimization algorithms of mathematical analysis and mathematical programming. Several quite typical calculation tasks are analysed, falling within the scope of subjects such as Mathematical Analysis and Mathematical Programming.



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Using Karush-Kuhn-Tucker conditions solve the nonlinear programming problem: maximize function $f(x,y) = x + 3y$, subject to: $9 - x^2 - y^2 \geq 0$.



ChatGPT

To solve the nonlinear programming problem using Karush-Kuhn-Tucker (KKT) conditions, let's first establish the components:

Objective Function:

$$f(x, y) = x + 3y$$

Constraint:

$$g(x, y) = 9 - x^2 - y^2 \geq 0$$

Scan1: Formulation of the optimization problem

The Lagrangian for this problem is:

$$L(x, y, \lambda) = f(x, y) + \lambda \cdot g(x, y) = x + 3y + \lambda(9 - x^2 - y^2)$$

KKT Conditions:

1. **Stationarity:**

$$\nabla L(x, y, \lambda) = 0$$

2. **Primal feasibility:**

$$g(x, y) \geq 0$$

3. **Dual feasibility:**

$$\lambda \geq 0$$

4. **Complementary slackness:**

$$\lambda \cdot g(x, y) = 0$$

Scan2: KKT Conditions

Compute Stationarity:

Solve the system of equations derived from the partial derivatives of L with respect to x , y , and λ .

Let's proceed to compute these derivatives and solve the resulting system of equations.

 Error analyzing ▾

The system of equations yields two sets of solutions for x , y , and λ :

1. $x = -\frac{3\sqrt{10}}{10}, y = -\frac{9\sqrt{10}}{10}, \lambda = -\frac{\sqrt{10}}{6}$
2. $x = \frac{3\sqrt{10}}{10}, y = \frac{9\sqrt{10}}{10}, \lambda = \frac{\sqrt{10}}{6}$

Again, the dual feasibility condition ($\lambda \geq 0$) allows only the second solution, as λ must be non-negative.

Thus, the optimal solution that maximizes the function $f(x, y) = x + 3y$ under the constraint $9 - x^2 - y^2 \geq 0$ is $x = \frac{3\sqrt{10}}{10}, y = \frac{9\sqrt{10}}{10}$.

Scan3: Solution of the system of equations of KKT Conditions

Verify with Complementary Slackness and Primal Feasibility:

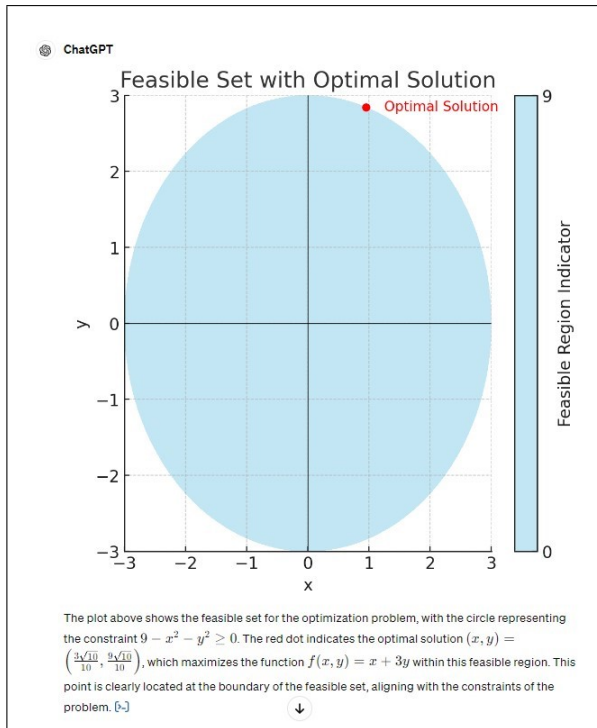
- Complementary Slackness: $\lambda \cdot g(x, y) = 0$

- Primal Feasibility: $g(x, y) \geq 0$

We can compute the constraint $g(x, y)$ with the chosen x and y to confirm these conditions. [↗]

The constraint function $g(x, y) = 9 - x^2 - y^2$ evaluates to 0 for the valid solution $x = \frac{3\sqrt{10}}{10}$, $y = \frac{9\sqrt{10}}{10}$, and the product of λ and $g(x, y)$ is also 0. This satisfies both the complementary slackness condition and the primal feasibility condition. Therefore, this solution indeed maximizes the function $f(x, y) = x + 3y$ under the given constraint, and all KKT conditions are satisfied. [↗]

Scan4: Verify with Complementary Slackness and Primal Feasibility



Scan5: Feasible Set with Optimal Solution

The obtained solution is correct and is given in symbolic form. When asked to Chat GPT: what computational program was used by the Chat to obtain a response in symbolic form, the answer was: a Python library for symbolic computations called SymPy was used. The plot were created by Chat GPT using Matplotlib, a widely used library in Python for generating static interactive and animated visualizations.

Keywords

Mathematical analysis, Mathematical programming, Chat GPT, Optimization methods, Computer Algebra Systems

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